

# Stevens-Johnson Syndrome Associated with Neuroval Administration in a Young Adult: A Severe Cutaneous Adverse Reaction with 8% Body Surface Area Detachment

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**ABSTRACT -- Background:** Neuroval (neuroal HCl) is a recently introduced centrally active sodium-channel modulator approved for neuropathic pain syndromes. Severe cutaneous adverse reactions (SCARs), including Stevens-Johnson syndrome (SJS), represent rare but catastrophic life-threatening complications. **Case Presentation:** A 24-year-old female initiated on Neuroval 150 mg daily for idiopathic trigeminal neuralgia presented 18 days later with sudden high fever (39.5 C), widespread purpuric macules with atypical targetoid morphology, epidermal detachment (~8% BSA), painful hemorrhagic crusting of lips, and severe bilateral purulent conjunctivitis. The patient was admitted to the intensive care burn unit. Histopathology confirmed full-thickness epidermal necrolysis pathognomonic of SJS. **Management & Outcome:** Neuroval was immediately withdrawn on presentation. Urgent multi-disciplinary intervention comprised high-dose intravenous immunoglobulin (IVIG 1 g/kg/day for 3 days), dedicated burn wound dressings, aggressive fluid replacement, and prophylactic amniotic membrane eye conformers. The patient achieved complete re-epithelialization by day 21 without long-term visual impairment or cutaneous scarring. **Keywords:** Stevens-Johnson Syndrome (SJS); Neuroval (neuroal HCl); Purpuric Macules with Atypical Targetoid Morphology; Epidermal Detachment (~8% BSA); Hemorrhagic Crusting of Lips; Bilateral Purulent Conjunctivitis; High Fever (39.5 C); Burn Unit Hospitalization; Pharmacovigilance.

## 1. INTRODUCTION

Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN) represent a severity spectrum of acute, life-threatening immunological drug reactions characterized by extensive apoptotic death of keratinocytes leading to epidermal sloughing and severe mucositis [1]. SJS is strictly defined as epidermal detachment affecting less than 10% of total body surface area (BSA), whereas TEN involves greater than 30% BSA [2]. Neuroval (neuroal hydrochloride) is a novel therapeutic agent targeting voltage-gated neural channels, recently indicated for refractory neuropathic pain syndromes including trigeminal neuralgia. We present a documented case of life-threatening SJS in a young female occurring 18 days after Neuroval initiation, requiring intensive care burn unit admission and specialized immunomodulatory salvage.

## 2. PATIENT PRESENTATION & CLINICAL SIGNS

A 24-year-old female presented to the emergency department with acute onset of high fever (39.5 C), severe chills, sore throat, painful burning eyes, and a rapidly spreading skin eruption. Her past medical history was notable only for idiopathic trigeminal neuralgia refractory to carbamazepine (discontinued 6 months prior due to mild somnolence). Eighteen days prior to the current admission, her neurologist prescribed Neuroval 150 mg PO daily. She was on no other prescription, over-the-counter, or herbal medications, and had no known drug allergies.

On examination, the patient was in acute toxic distress. Vital signs: temperature 39.5 C, blood pressure 104/62 mmHg, pulse 124 bpm, respiratory rate 20 breaths/min, and oxygen saturation 98% on ambient air. Dermatological examination revealed extensive, confluent purpuric macules with atypical targetoid morphology distributed over the face, neck, trunk, and proximal extremities. Flaccid bullae and sheets of epidermal detachment (~8% BSA) were present on the chest and back, with a positive Nikolsky sign. Mucosal examination revealed exquisite tenderness, extensive denudation, and hemorrhagic crusting of the lips and oral mucosa. Ophthalmic evaluation demonstrated bilateral purulent conjunctivitis with marked chemosis, photophobia, and early pseudomembrane formation. Genital exam showed superficial mucosal erosions.

## 3. DIAGNOSTIC WORKUP & HISTOLOGY

Admission laboratory workup showed mild leukopenia (WBC 3.4 x 10<sup>9</sup>/L), elevated C-reactive protein (84 mg/L), normal serum creatinine (0.8 mg/dL), and blood urea nitrogen (14 mg/dL). Serum transaminases and electrolytes were within normal reference ranges. Blood, sputum, and conjunctival cultures showed no bacterial growth; viral PCR testing for HSV-1/2, VZV, and Mycoplasma pneumoniae was negative. The calculated SCORTEN severity score on admission was 2 (heart rate >120 bpm and detached BSA <10%), corresponding to a predicted mortality risk of 12.1%.

An urgent 4-mm punch skin biopsy from the margin of a progressive blister revealed extensive full-thickness epidermal necrosis, widespread basilar keratinocyte apoptosis, subepidermal blister formation with cell-poor separation, and sparse superficial perivascular lymphocytic infiltration. Direct immunofluorescence (DIF) was entirely negative for immunoglobulin or complement deposition, definitively excluding autoimmune bullous diseases.

Table 1. Clinical Timeline, SCORTEN Parameters, and Mucocutaneous Evolution

| Timeline                 | Clinical Event / Intervention                                          | Detachment / Status                       |
|--------------------------|------------------------------------------------------------------------|-------------------------------------------|
| Day 0                    | Neuroval 150 mg daily started for trigeminal neuralgia                 | Baseline (0% BSA)                         |
| Day 18                   | Fever 39.5 C, targetoid macules, lip crusting, purulent eye discharge  | <b>8% BSA detachment</b> (SCORTEN = 2)    |
| Day 19-21                | ICU Burn Unit admission; IVIG (1 g/kg/day x 3 days); amniotic membrane | Detachment arrested at ~8% BSA            |
| Day 28                   | Ward transfer; oral crusting sloughing; active re-epithelialization    | ~70% re-epithelialized                    |
| Day 39 (Day 21 post-adm) | Full clinical recovery; ophthalmology clearance; discharged            | <b>100% complete re-epithelialization</b> |

## 4. ICU BURN UNIT MANAGEMENT & OUTCOME

Due to the aggressive, life-threatening progression of epidermal detachment and multisite mucosal necrosis, the patient was admitted to the intensive care burn unit. Neuroval was discontinued immediately on presentation (positive dechallenge). Supportive care included a warm ambient environment (30-32 C), strict aseptic handling, careful fluid resuscitation based on urinary output, nutritional supplementation, and non-adherent silicone foam barrier dressings without aggressive debridement. Intravenous immunoglobulin (IVIG) was administered at a total dose of 3 g/kg divided over three consecutive days (1 g/kg/day on days 19, 20, and 21) to halt Fas-mediated keratinocyte apoptosis [3].

Ophthalmology placed temporary amniotic membrane conformers to prevent symblepharon and corneal scarring. The progression of skin detachment halted by day 20. Re-epithelialization began on day 24, and by day 39 (day 21 post-admission), complete re-epithelialization (~100%) was achieved across all denuded surfaces. Mucosal erosions resolved, and visual acuity remained intact (20/20 bilaterally). The patient was discharged in stable condition. Re-challenge with Neuroval is strictly contraindicated and was permanently documented in her health record.

## 5. CAUSALITY & REGULATORY IMPLICATIONS

Causality assessment utilizing the Algorithm of Drug Causality for Epidermal Necrolysis (ALDEN) scored 7 points, confirming a "Definite/Probable" causal relationship between Neuroval and SJS. The 18-day latency precisely reflects the typical 1-4 week exposure window required for drug-antigen sensitization and cytotoxic CD8+ T-cell expansion in SCARs [4]. Under international pharmacovigilance regulations (ICH E2D, FDA 21 CFR 314.80), this event fulfills multiple seriousness criteria: **Life-Threatening**, requiring **Inpatient Hospitalization** and ICU burn unit care. Immediate regulatory ICSR reporting was executed.

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